

Shanghai Jiao Tong University  
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Shanghai, China

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## Research Overview

My undergraduate research focused on computer vision, with an emphasis on visual recognition and computational photography. During my master's studies, my research has centered on embodied intelligence, including zero-shot navigation, robotic manipulation, and post-training for vision-language-action models. My current work investigates the integration of world models and navigation, with particular emphasis on the relationship between learned world-model representations and navigational maps. In doctoral study, I aim to further develop world models and embodied agent systems for robust perception, planning, and interaction in physical environments.

**Research Areas:** Embodied Policies and Agent Systems, World Models and Vision Foundation Models, Reinforcement Learning Algorithms and Systems.

## Education

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|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M.Eng.</b><br>Sep 2024–Mar 2027<br>(Expected) | <b>Shanghai Jiao Tong University (SJTU)</b><br>SJTU Paris Elite Institute of Technology (SPEIT)<br>Electronic Information<br>Shanghai, China.                   |
| <b>Exchange Student</b><br>Sep 2024–Feb 2025     | <b>Institut National des Sciences Appliquées de Lyon (INSA Lyon)</b><br>Department of Computer Science and Information Technology (IF)<br>Lyon, France.         |
| <b>B.Eng.</b><br>Sep 2020–Jun 2024               | <b>Shanghai Jiao Tong University (SJTU)</b><br>SJTU Paris Elite Institute of Technology (SPEIT)<br>French; Minor in Information Engineering<br>Shanghai, China. |

## Research and Industry Experience

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|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Research Intern</b><br>Mar–Jul 2026 | <b>Agibot Robotics</b> <ul style="list-style-type: none"><li>• Worked on RL post-training for the Longcheer phone-packaging project.</li><li>• Investigated whole-body control for wheeled-legged robots, including waist-and-chassis data collection and relative-position control to improve foothold accuracy.</li></ul> |
| <b>Research Intern</b><br>Jun–Sep 2024 | <b>Alibaba / Alibaba Cloud</b><br>Studied visual navigation and goal-conditioned representation learning, with a focus on ViNT and NoMaD-style methods for learning navigation policies from visual observations.                                                                                                           |
| <b>Product Intern</b><br>Jun–Aug 2022  | <b>Alibaba Cloud</b><br>Gathered partner feedback and refined product proposals for the China Academy of Art low-carbon platform; prepared a requirements document for product-manager review. Supported carbon-footprint assessment and campus analytics.                                                                  |

## Selected Projects

### Industry Projects

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|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main Contributor</b><br>Mar–Jul 2026 | <b>Longcheer Phone Packaging via Real-Robot RL Post-Training</b><br>Developed and benchmarked DAGger, AWR, and ALOE for real-robot RL post-training on Agibot's G02 platform.<br>Achieved 90% success on the long-horizon Longcheer phone-packaging task, establishing a repeatable evaluation pipeline for mobile manipulation.                                                                     |
| <b>Main Contributor</b><br>Jun–Aug 2022 | <b>Product Carbon Footprint and Low-Carbon Operations, Alibaba Cloud</b><br>Built a product life-cycle assessment workflow from process data and emission factors in CPCD-China and Ecoinvent.<br>Delivered the carbon-footprint certification report for the Hangzhou Asian Games mascot plush toy, the Games' first zero-carbon licensed product, and produced weekly low-carbon campus analytics. |

## Academic Projects

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|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Undergraduate Innovation Practice Program (IPP)</b><br>Sep 2022–Jun 2024 | <b>Open-World Person Tracking and Identity Recognition</b><br>Developed a working prototype for person detection, tracking, face recognition, and video retrieval in unconstrained surveillance scenes.<br>Reproduced RetinaFace with ResNet-50 and YOLOv5 detection, and established initial gait-face alignment and multimodal-fusion baselines.     |
| <b>Participation in Research Program (PRP)</b><br>Jul 2021–Jan 2022         | <b>Energy Internet Platform Data Interface Design and Development</b><br>Developed data interfaces for an integrated energy demonstration system in a smart-grid building.<br>Built Schneider Electric-based control and visualization modules that ingested and displayed real-time energy-operation data, delivering a working monitoring prototype. |

## Publications

The \* symbol denotes equal contribution, and the † symbol denotes corresponding authors.

### Conference Papers

- C1 Yijin Zhou, Linqian Zeng, Xiaoya Lu, **Wenyuan Xie**, Dongrui Liu, Junchi Yan, Jing Shao. “Exploring Agentic Tool-Calling Decisions via Uncertainty-Aligned Reinforcement Learning.” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026. [arXiv:2606.06976](#).
- C2 Guodong Zhang, Qichen He, **Wenyuan Xie**, Shaokai Wu, Yanbiao Ji, Qiuchang Li, Bayram Bayramli, Yue Ding, Hongtao Lu. “RelAfford6D: Relational 6D Affordance Graphs for Constraint-Driven Robotic Manipulation.” *European Conference on Computer Vision (ECCV)*, 2026. [arXiv:2606.27036](#).
- C3 Yanbiao Ji, Qiuchang Li, Yuting Hu, Shaokai Wu, **Wenyuan Xie**, Guodong Zhang, Qicheng He, Deyi Ji, Yue Ding, Hongtao Lu. “Recovering Hidden Reward in Diffusion-Based Policies.” *International Conference on Machine Learning (ICML)*, 2026. [arXiv:2605.00623](#).
- C4 Yijin Zhou, Yutang Ge, **Wenyuan Xie**, Linqian Zeng, Xiaowen Dong, Yuguang Wang. “How Out-of-Distribution Detection Learning Theory Enhances Transformer: Learnability and Reliability.” *International Conference on Machine Learning (ICML)*, 2026. [arXiv:2406.12915](#).
- C5 **Wenyuan Xie**, Shaokai Wu, Yijin Zhou, Yanbiao Ji, Guodong Zhang, Bayram Bayramli, Qiuchang Li, Xunchu Zhou, Yue Ding, Hongtao Lu. “MVP-Nav: Multi-layer Value Map Planner Navigator.” *Robotics: Science and Systems (RSS)*, 2026. [arXiv:2606.31919](#).
- C6 Shaokai Wu, Yanbiao Ji, Qiuchang Li, Zhiyi Zhang, Qichen He, **Wenyuan Xie**, Guodong Zhang, Bayram Bayramli, Yue Ding†, Hongtao Lu†. “Dejavu: Towards Experience Feedback Learning for Embodied Intelligence.” *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [arXiv:2510.10181](#).

### Preprints

- P1 **Wenyuan Xie**, Mengyang Hong, Yongzhong Wang, Yanbiao Ji, Yijin Zhou, Shaokai Wu, Shalayiding Sirejiding, Huayi Zhou, Yi-Chao Chen, Ma Ling, Yue Ding, Hongtao Lu. “Navi-Agent: Unlocalized Monocular Navigation Agent.” *arXiv preprint*, 2026. [arXiv:2609.20388](#).
- P2 Yanbiao Ji, Qiuchang Li, Shaokai Wu, **Wenyuan Xie**, Deyi Ji, Qi Zhu, Yue Ding, Hongtao Lu. “Capability Loss in Embodied Recursive Self-Improvement.” *International Conference on Learning Representations (ICLR)*, 2027, under review. [OpenReview: IkYmTAyPXZ](#).

## Awards and Honors

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|--------------------------------------------------------------------|----------------------------------------|
| <b>SPEIT Outstanding Freshman Scholarship</b>                      | 2020–2021; CNY 10,000.                 |
| <b>SJTU Undergraduate Academic Excellence Scholarship</b>          | 2022–2023, 2023–2024; CNY 1,000 each.  |
| <b>SJTU Graduate Academic Scholarship</b>                          | 2024–2025; CNY 7,800.                  |
| <b>SJTU Basketball Sports Association Cup</b>                      | 2020–2021, 2022–2023; Quarterfinalist. |
| <b>1st Zhiyuan Cup Graduate Men’s 3-on-3 Basketball Tournament</b> | 2025; Third Place.                     |

## Technical Skills

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|------------------------------|---------------------------------------------------------------|
| <b>Programming Languages</b> | C++, Java, Python, PyTorch, Git.<br>Chinese, English, French. |
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